H

millennia institute

Class: Adm No

Candidate Name: _____

2026 Preliminary Examinations
Pre-University 3

GEOGRAPHY (HIGHER 2) **9173/01**
Paper 1 **31 August 2026**

3 hours

Additional Materials: Insert, Answer Booklet(s)

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front of the answer booklet. If you need additional writing paper, ask the invigilator for a continuation booklet.

Section A
Answer **both** questions.

Section B
Answer **two** questions, **one** from each cluster.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer **both** questions.

Cluster 1 Development, Economy and Environment

1 To date, BYD achieved record sales of 8.2 million vehicles globally. Resource 1 shows an BYD's Global Production Network (GPN) across China, Southeast Asia, Europe and the Americas. Resource 2 shows a photograph of BYD's Mega Base in Zheng Zhou, China. Resource 3 shows an infographic depicting the additional tariffs applied to newly imported Chinese electric vehicles to the European Union in 2024 by the European Commission. Resource 4 shows the key minerals used in producing an electric vehicle battery. Resource 5 shows an infographic of the world's largest dams and their impacts.

a)	Explain the challenges limiting the progress towards sustainable development.	[4]
	Indicative Content • Economic challenges ○ Measures to reduce environmental impacts may require **costly technologies and infrastructure**, such as renewable energy systems, wastewater treatment facilities and cleaner production technologies. [1] These impose high **capital and maintenance costs**, particularly where governments and firms have limited financial and technological capacity. [+1] • Political challenges ○ Governments may prioritise **short-term national interests** over longer-term environmental goals, while countries may disagree over who should bear the costs of addressing global environmental problems. [1] ○ This can delay or weaken the implementation of **international agreements and coordinated action to achieve progress towards SD**. [+1]	
b)	Using Resource 1, describe the ways by which BYD's GPN connects places within the global economy through flows of materials and capital.	[6]
	Indicative Content BYD's GPN connects places through the **movement of raw materials, semi-processed components and finished vehicles** between different production and consumption sites. [1] For example, materials and components such as batteries and electronic parts are sourced from suppliers in different countries and transported to BYD's production facilities in Szeged, Rayong and Camacari, where they are subsequently assembled into vehicles. [+1] OR Finished vehicles are then transported from production locations to overseas consumer markets for sale. [+1] BYD's GPN also connects places through **flows of capital between China and overseas host economies**. BYD invests capital in establishing and expanding production facilities overseas to capitalise on comparative advantage, [1] while revenue generated from the sale of vehicles in overseas markets flows back to the company. [+1] For example, BYD has invested in overseas manufacturing facilities	

	such as its electric-vehicle plant in **Hungary**, linking Chinese corporate capital with investment in European production. [+1] OR BYD's GPN connects places through the **flow of materials** between different locations involved in the production of electric vehicles. This connects resource-producing and component-producing places to BYD's assembly and production locations. [1] For example, **raw materials and components such as lithium-ion batteries and electronic components** are sourced from different countries [+1] and transported to BYD's manufacturing facilities in China and overseas. [+1] BYD's GPN also connects places through **flows of capital**. BYD invests capital in overseas production facilities, such as its manufacturing operations in **Hungary and Thailand**, allowing production to take place closer to overseas markets. [1] Capital therefore flows from BYD and its investors into different locations, while investment in these places supports production and strengthens their links to BYD's global production network. [+1] HQ → advanced R&D Centers → EV Manufacturing plants and clusters [+1] Thus, BYD's GPN connects places through **interdependent flows of materials and capital**, linking locations involved in resource extraction, component production, manufacturing and markets. [+1]	
c)	With reference to Resources 1 and 2, explain positive and negative impacts that GPNs of TNCs, like BYD, on host economies.	[6]
	Indicative Contents **Positive impacts** 1. **Employment and incomes** TNCs such as BYD can establish production facilities in host economies, creating large-scale, **direct employment** in manufacturing and **indirect employment** through local suppliers, forward and backward production linkages and services. [1] This generates household incomes and can contribute to economic growth, serving to diversify the local economy as well. [+1] 2. **Capital investment and infrastructure** Foreign direct investment (FDI) from TNCs brings **capital into the host economy** through corporate taxations, purchase of materials from local suppliers and establishment of supporting operations. [1] These FDI can result in better telecommunications and road networks, and other supporting infrastructure can increase productive capacity and stimulate ripple effects in the national or local economic activity. [+1] 3. **Technology and skills transfer** Local workers and firms may gain access to **new technologies, production techniques and managerial expertise**, [1] potentially improving productivity and enabling domestic firms to participate more deeply in GPNs. [+1] 4. **Trade and market access** Integration into a TNC's GPN can increase the host economy's **exports and participation in global trade**, [1] while local suppliers may gain access to international markets through the TNC's GPN. [+1]	

	Negative impacts 1. **Dependence on TNCs** Host economies may become dependent on foreign firms for employment, investment and exports. [1] If the TNC relocates production because of changing costs or market conditions, the host economy may experience **job losses and reduced economic activity**. [+1] 2. **Limited value capture** Although production occurs in the host economy, a significant proportion of the **profits and higher-value activities** such as R&D and strategic decision-making may remain with the TNC's headquarters elsewhere. [1] The host economy therefore may capture only part of the value generated, while most of the profits are repatriated back to the TNC's home economies. [+1] 3. **Pressure on local firms** Large TNCs can compete directly with domestic firms from within the same industry, potentially causing smaller local producers to lose market share or become dependent on the TNC as suppliers. [1] 4. **Environmental and social costs** Rapid expansion of manufacturing can increase **resource consumption, pollution, waste and pressure on land and infrastructure**, particularly where environmental regulation is weaker. [1] Each point and well-explained reason will be awarded 1m, up to a maximum of 5m. A maximum of 3m is awarded if response only indicates positive **or** negative impact.	
d)	Using Resources 2 and 3, explain the influence of states and multilateral institutions on TNCs' operations.	[4]
	Indicative Contents Resource 2 shows BYD's **large-scale manufacturing base in Zhengzhou, China**, illustrating how the Chinese state can create conditions that enable TNCs to expand their production through strengthening their production capacity and achieve economies of scale. [1] State support can include **infrastructure provision (e.g. connectivity to transport hubs), industrial policies, investment incentives and support for strategic industries**. [+1] Resource 3 shows that the **EU imposed different tariffs on Chinese-made EVs depending on the companies' cooperation with the EU anti-subsidy investigation**. [1] In the final measures, BYD was subject to a **17.4% EU duty**, while non-cooperating companies faced a **38.1% duty to influence foreign TNCs to align with regional regulations**. [+1] This demonstrates that states can **alter the costs and market access of TNCs through trade policies**. Higher tariffs increase the cost of exporting into the EU, potentially affecting TNC's pricing, competitiveness and decisions about where to locate production. [+1] Thus, government policies can influence **TNCs' GPNs and investment decisions**, rather than TNCs operating independently of states. Award 1m for each relevant point using information as shown in R2 or R3.	

e)	With reference to Resource 4, suggest the economic and environmental impacts that extractive industries have on places.	[4]
	Environmental Impacts • **Land degradation and habitat loss:** Extraction of minerals such as **lithium, cobalt, nickel and manganese** can involve open-pit or other large-scale mining, resulting in **vegetation clearance, soil disturbance and habitat destruction**. [1] ○ This can reduce biodiversity and alter ecosystems, as mineral extraction and processing may lead to **water depletion and pollution,** [+1] while mining waste can contaminate surface and groundwater, reducing water quality for ecosystems and human use. [+1] • **Air pollution:** Mining, crushing and transporting minerals can generate **dust and particulate matter**, [1] ○ while the use of heavy machinery and processing facilities produces carbon and pollutive gaseous emissions.[+1] This can reduce local air quality and affect surrounding ecosystems. • **Irreversible structural changes to natural landforms and landscapes:** Extractive industries can irreversibly alter environments by physically removing and reshaping existing landforms during mining. [1] ○ The removal of rock and soil, together with habitat destruction and changes to drainage and groundwater flows, can fundamentally alter the physical and ecological characteristics of a place. [+1] Even after mine closure and rehabilitation, the original landform and ecosystem may therefore not be fully restored. **Economic Impacts** • **Extraction creates direct employment in EIs**, like mining and processing, while generating indirect employment in transport, equipment supply and supporting services locally. [1] ○ This can raise household incomes and stimulate local economic activity. [+1] • **EIs often breed dependence on a single extractive commodity or export market** can make places vulnerable to commodity-price fluctuations and changes in external demand. [1] ○ Heavy dependence on mineral extraction may also discourage diversification of the economy. [+1]	
f)	Explain the limitations and trade-offs of alternative energy sources in providing energy security as shown in Resources 5.	[6]
	Indicative Content **1. Large generating capacity, but geographical limitations** Hydroelectric dams can generate large quantities of electricity, as shown by the very high capacities of major dams. This can reduce dependence on imported fossil fuels and improve **energy security**. [1] However, suitable sites require **large rivers, adequate water flow and appropriate topography**, so hydropower cannot be developed equally in all countries. [+1] **Trade-off:** Countries with suitable physical geography may gain substantial domestic electricity generation but result in knock-on social and environmental impact on the displaced riparian communities and downstream water quality and quantity. [1]	

	2. Renewable, but dependent on water availability Hydropower is renewable because the water cycle continually replenishes the resource. However, electricity generation depends on **river discharge and reservoir levels**. [1] Droughts and changing precipitation patterns can reduce water availability and therefore electricity generation. [+1] **Trade-off:** Hydropower reduces dependence on finite fossil fuels, but its substantial financial cost can make large-scale hydropower difficult for poorer countries to develop, or dependent on financial loans and reliance on foreign companies to drive such projects. [1] Maximum of 3m awarded if either limitations or trade-offs is not explained.	

Cluster 2 Tropical Environments

2	Myanmar is a country in the tropical region. Resource 6 shows a climograph of Thaton, Mon State in Myanmar. Resource 7 shows the rainfall patterns recorded at Thaton, Hpa-an and Mawlaymyine between 4 to 9 August 2019. Resource 8 shows an aerial photograph of a mass movement event in Thalphyugone Village at Ma Latt hill, in Paung Township, Mon State on 9 August 2019, located upstream of the Gyaing River. Resource 9 shows a satellite image of the Gyaing River in Mon State, Myanmar.		
	(a)	Cite data from Resource 6 to describe the temperature and rainfall characteristics of Thaton, Mon State, Myanmar.	[4]
		• *Award 1 mark for each valid point, up to a maximum of 4 marks.* • *Responses must cite relevant data from Resource 1 where appropriate.* Acceptable responses include: • Tropical monsoon climate (Am). [1] + mention distinct wet and dry season. • **High** mean annual temperature of 26.4°C. [1] • **Small** annual temperature range of approximately 4.6°C (about 24.8°C to 29.4°C). [1] • **High** total annual precipitation of 3922 mm. [1] • **Distinct wet and dry seasons**, strong seasonal rainfall pattern where it is much wetter from May to October and much drier from November to April; e.g. rainfall peaks at 906 mm in July, compared with only 3 mm in February. [1]	
	(b)	With reference to Resource 6, account for the seasonal rainfall pattern experienced in location such as Myanmar.	[6]
		• *Award 1 mark for each valid point, up to a maximum of 6 marks* • During the Northern Hemisphere summer, the sun is overhead the Tropic of Cancer, which leads to strong heating of the Asian landmass that causes air to rise, creating an area of low pressure over Asia. [1] • In the Southern Hemisphere, further away from the sun, experiences winter, creating an area of high pressure. [1] • Due to pressure gradient force, wind moves from the higher-pressure Southern Hemisphere towards the low pressure over Asia. [1] • As the wind crosses the Equator, the winds are deflected to the right by the Coriolis effect, forming the south-west monsoon winds. [1] • The south-west monsoon travels across the Indian Ocean/Bay of Bengal, picking up large amounts of moisture and bringing heavy rainfall to Myanmar, especially around June–August, with rainfall peaking in July in Resource 1. [1] • However, during the Northern Hemisphere winter, the Asian landmass cools rapidly, producing high pressure over continental Asia; winds therefore reverse to form the north-east monsoon. [1] • These north-east monsoon winds originate mainly over the Asian landmass and contain much less moisture, resulting in the low rainfall/dry season from approximately November to April shown in Resource 1. [1]	
	(c)	Using Resource 7, compare the rainfall patterns recorded at Thaton, Hpa-an and Mawlamyine between 4 and 9 August 2019.	[4]

		• *Award 1 mark for each valid comparison (1 similarity; 1 difference) supported by data, up to a maximum of 4 marks.* • *Accept other valid similarities, differences or anomalies. Comparisons should be supported by appropriate data from the resource.* • *Accept reasonable estimates from the graph.* [General Trend] Accumulated rainfall increased at all three stations between 4 and 9 August, from about 50–60 mm initially to between 440 and 770 mm. [1] [Similarity] • Thaton and Hpa-an showed very similar rainfall patterns, both recording their highest daily rainfall on 5–6 August, at about 165 mm and 160 mm respectively, and reaching similar accumulated totals of about 450 mm and 440 mm. [1] • Similar starting point, where all three stations recorded relatively similar rainfall on 4–5 August, at approximately 45–60 mm. [1] [Difference] • On 5–6 August, Thaton and Hpa-an recorded about 160–165 mm, considerably higher than Mawlamyine at about 100 mm. [1] • Mawlamyine received much greater accumulated rainfall overall, reaching about 770 mm by 9 August, compared with approximately 450 mm at Thaton and 440 mm at Hpa-an. [1] • Mawlamyine's maximum daily rainfall (approximately 355 mm) was more than twice the maximum daily rainfall recorded at either Thaton or Hpa-an (approximately 160–165 mm). [1] [Anomaly] Mawlamyine experienced an exceptionally large rainfall event on 8–9 August, receiving about 355 mm in one day, compared with only about 75 mm at Thaton and 95 mm at Hpa-an. [1]	
	(d)	Using Resources 6, 7 and 8, suggest how various factors may have contributed to the mass movement event as shown in Resource 8.	[6]
		• *Award 1 mark for identifying and explaining the factor.* • *Award an additional 1 mark for explaining how it changes shear stress or shear strength, leading to mass movement.* • *Accept other valid explanations based on the resources.* Acceptable responses include: • [Resource 6/7] With heavy monsoonal rainfall, the area experiences very high rainfall during the monsoon, with rainfall reaching about 906 mm in July. [1] ○ The rainfall saturates the slope material, increasing its weight and pore-water pressure; this increases shear stress and reduces shear strength, allowing saturated material to move downslope as a mud/debris flow. [+1] • [Resource 8] Dense vegetation is present on the slope, adding additional mass/weight to the slope material. [1]	

		○ The additional weight increases the downslope force exerted by gravity and therefore increases shear stress; if shear stress exceeds the slope's shear strength, the material fails and moves downslope. [+1] • [Resource 8] The photograph shows the affected area associated with a sloping terrain, with steep topography. [1] ○ A steeper slope increases the downslope component of gravity, thereby increasing shear stress; when this exceeds the shear strength of the slope material, mass movement occurs and material can move rapidly downslope. [+1]	
	(e)	Using Resources 6 and 8, explain the formation of the fluvial landform as shown in Resource 9.	[6]
		• *Award **1 mark for each valid point**, up to a maximum of **6 marks.*** • *Accept other valid explanations. Credit appropriate use of evidence from the resources.* Acceptable responses include: • The fluvial landform shown is a braided river (no marks awarded for this) • During the monsoon season, high rainfall produces periods of high river discharge (Resource 6). The increased velocity and discharge increase the river's competence and capacity, increasing erosion [1] ○ allowing large quantities of sediment supplied by the mass movement in Resource 7 to be transported downstream. [1] • During the dry season, lower rainfall results in lower river discharge (Resource 6). The river's competence and capacity decrease, causing coarser sediment to be deposited first, [1] ○ followed by accumulation of finer sediment, and repeated deposition builds mid-channel bars. [1] • The mid-channel bars shown in Resource 9 obstruct the river flow, forcing water to divert around them into multiple channels that split and rejoin, producing a braided channel pattern. [1] • Over time, vegetation may colonise and stabilise some bars, forming more permanent islands. [1] • The seasonal alternation between high and low discharge repeats the processes of erosion, transportation and deposition, maintaining and modifying the braided channel over time. [+1]	
	(f)	Explain the differences between physical and chemical weathering processes.	[4]
		• *Award 1 mark for explaining each weathering process* • *Award additional 1 mark for a relevant example or elaboration, up to a maximum of 4 marks.* • *Accept other valid examples or elaborations.* Acceptable responses include: • Physical weathering involves the breakdown or disintegration of rock into smaller fragments without changing its chemical composition. [1] ○ It occurs through mechanical processes such as pressure release, freeze-thaw, thermal expansion or salt crystallisation, where physical stresses cause rock to fracture. [+1]	

		○ For example, freeze-thaw weathering occurs when water enters cracks, freezes and expands, exerting pressure that widens the cracks and eventually breaks the rock apart. [+1] • Chemical weathering involves the decomposition or alteration of rock minerals through chemical reactions, changing the rock's mineral/chemical composition. [1] ○ Processes such as carbonation, hydrolysis and oxidation react with minerals in the rock, causing minerals to dissolve or be transformed into new minerals/materials. [+1] ○ For example, carbonation occurs when carbon dioxide dissolves in rainwater to form weak carbonic acid, which reacts with and dissolves calcium carbonate in limestone. [+1]	

Section B

Answer **either** Question 3 **or** Question 4, and **either** Question 5 **or** Question 6.

	Cluster 1 Development, Economy and Environment	
3	Evaluate the relative importance of ecosystem services provided by the environment.	[20]
	Indicative Content *Candidates should compare the **relative importance** of provisioning, regulating, cultural and supporting ecosystem services rather than merely describing them. Importance may be assessed by their contribution to human survival and well-being, scale of benefits, consequences of their loss, substitutability and dependence of other services on them. Provisioning services meet immediate needs such as food and water; regulating services reduce environmental risks; supporting services sustain other ecosystem services; and cultural services contribute to recreation, identity and well-being. Relevant examples should support these comparisons.* *Higher-level responses should recognise that no universal ranking applies. The relative importance of each service varies by stakeholder, geographical context, spatial scale and timescale. Provisioning services may be most important to communities directly dependent on natural resources, while regulating services may dominate in hazard-prone environments. Supporting services may be most fundamental in the long term because other services depend on them. Strong responses should also evaluate interdependence and trade-offs between services before reaching a qualified judgement. The strongest conclusions will therefore offer a reasoned and qualified judgement: supporting or regulating services may provide the most fundamental basis for long-term human well-being, but the relative importance assigned to each service ultimately depends on the evaluative criterion and the needs, vulnerabilities and priorities of different beneficiaries.*	

4	"The Global Production Networks (GPNs) of Transnational Corporations (TNCs) contribute to the uneven development of home and host economies." Evaluate this statement.	[20]
	Indicative content *Students should understand that TNCs organise GPNs through the spatial division of different economic activities across home and host economies, resulting in uneven opportunities to capture value. Higher-value functions such as headquarters, R&D and management may remain concentrated in home economies, while lower-value manufacturing and assembly are located in lower-cost host economies. This can reinforce uneven development through differences in wages, skills development and value captured. However, host economies may also benefit from employment, export earnings, technology transfer and linkages with local firms, potentially reducing uneven development.* *Students should also understand that the impacts of GPNs are spatially differentiated and context-dependent, rather than inherently producing uneven development. The extent to which home and host economies benefit depends on their position within the GPN, bargaining power, level of development and degree of TNC embeddedness. Host economies with stronger institutions, skilled labour and policies promoting local linkages may capture greater value and upgrade into higher-value activities, while weaker economies may remain dependent on low-value functions. Hence, students should recognise that GPNs can contribute to uneven development, but outcomes depend on power relations and the capacity of economies to capture and retain value from their participation in GPNs.*	

	Cluster 2 Tropical Environments	
5	Evaluate the extent to which humans can control the occurrence of fluvial floods in the humid tropics.	[20]
	Indicative content *Students should understand that humans can control the occurrence of fluvial floods to some extent by modifying drainage basin processes and river discharge. Structural measures such as dams, channelisation and flood-storage schemes can regulate river flow, increase channel capacity or temporarily store excess water. Land-use management, including afforestation and reducing impermeable surfaces, can also increase interception and infiltration and reduce surface runoff. Their effectiveness varies with the scale of intervention, characteristics of the drainage basin and magnitude and frequency of rainfall events.* *Students should also understand that human control is constrained by the physical processes characteristic of the humid tropics. High and intense rainfall, saturated soils and extreme rainfall events can generate discharge that exceeds the design capacity of flood-management measures. Human activities such as urbanisation and deforestation may further increase runoff and flood occurrence. Hence, humans have greater ability to modify flood frequency and magnitude than completely prevent floods. The extent of control depends on interactions between human interventions and physical conditions, with control generally greater for smaller, more frequent floods than for extreme events that exceed infrastructural or drainage-basin thresholds.*	
6	Evaluate the extent of human activities in influencing the formation of karst landforms in the humid tropics.	[20]

	Indicative Content *Candidates should explain how natural processes and factors, particularly carbonation and solution, high rainfall, vegetation and soil CO_2, limestone lithology, geological structure and tectonic history, combine to form cockpit karst, tower karst and caves in the humid tropics. At the same time, provide a paragraph by paragraph comparison and assessment of how far human activities, such as quarrying, deforestation, agriculture, construction and tourism, can alter these processes or modify existing karst landforms.* *Stronger responses should use process-based explanations rather than simply describing impacts, and compare the relative importance, spatial scale and timescale of human and physical factors. Relevant examples could include Gunung Mulu (Malaysia), Phong Nha–Ke Bang (Vietnam), Guilin (China) and Ha Long Bay (Vietnam). Statistics should be used selectively to support geographical arguments, rather than as isolated facts. A clear judgement should recognise that human activities can have significant local impacts on existing karst, particularly through quarrying and development, but the fundamental formation and overall morphology of major tropical karst landscapes are generally more strongly controlled by natural processes operating over much longer timescales.*	

End of Paper

Copyright Acknowledgements

Resource 1 https://chinesecars.asia/byd-sales-record-2026/ (Last accessed 4 August 2026)
Resource 2 https://www.metro-society.com/en/leisure/byd-from-smart-factory-to-test-track-inside-byd-zhengzhou (Last accessed 4 Aug 2026)
Resource 3 https://source.benchmarkminerals.com/?camefrom=rhomotion.com (Last accessed 4 Aug 2026)
Resource 4 https://mcecleanenergy.org/ask-an-expert-whats-the-environmental-impact-of-ev-batteries/ (Last accessed 14 August 2026)
Resource 5 https://elements.visualcapitalist.com/visualizing-the-worlds-largest-hydroelectric-dams/ (Last accessed on 17 August 2026)
Resource 6 https://en.climate-data.org/asia/myanmar/mon/thaton-312/ (Last accessed on 1 August 2026)
Resource 7 https://link.springer.com/article/10.1007/s10346-021-01758-7 (Last accessed on 1 August 2026)
Resource 8 https://www.bnionline.net/en/news/mon-monsoon-landslide-kills-23-dozens-injured (Last accessed 1 August 2026)
Resource 9 Author's own https://earth.google.com/earth/d/1OwGu-Qoqf6YKFFc7HPXWSXV-LZXpmfsG?usp=sharing (Last accessed 1 August 2026)